DATA ANALYTICS PORTFOLIO PROJECT

HR Analytics Employee Attrition Dashboard Report

Understanding who leaves, and why, across 1,470 employees
Built with Excel and Power BI (DAX)

Prepared by

Syed Ubedullah Basha

Data Analyst | Power BI Developer | SQL Developer

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syedubedullahbasha@gmail.com | linkedin.com/in/syed-ubedullah-basha | github.com/syed-ubedullah

Executive Summary

What the data says, in one page

16.1%	30.5%	55.9%	29.8%
Overall attrition 237 of 1,470 employees	Overtime employees (vs 10.4% without)	Single + overtime + travel (34 employees)	Attrition in the first 2 years of tenure

This report presents a five-page Power BI dashboard built on the IBM HR Analytics Employee Attrition dataset. The goal is to understand which employees are most likely to leave and which factors are most strongly associated with attrition, so that HR teams can focus retention efforts where they matter most. Overall, 237 of 1,470 employees (16.1%) left the company.

Headline findings

- **Overtime has the strongest link to attrition.** Employees working overtime leave at 30.5% versus 10.4% without overtime, and they account for 54% of all attrition while making up only 28% of the workforce.
- **The first two years are the riskiest.** Employees with 0-2 years of tenure leave at 29.8%, versus 8.1% for those with 10+ years. Employees aged 18-25 leave at 35.8%.
- **Role and pay matter.** Sales Representatives have the highest attrition (39.8%) and the lowest average pay ($2.6K per month). Employees who left earned about 30% less than those who stayed ($4,787 vs $6,833).
- **Travel, work-life balance and stock options all show clear gaps.** Frequent travelers leave at 24.9% (vs 8.0% for non-travelers), bad work-life balance reaches 31.3%, and employees with no stock options leave at 24.4% (vs 9.4% at level 1).
- **Salary hike percentage alone does not separate leavers from stayers.** Attrition stays between 14.6% and 17.5% across hike bands.

How to read this report

Section 1 describes the project and data. Section 2 explains the preparation and model. Section 3 walks through each of the five dashboard pages with screenshots. Section 4 consolidates insights and recommendations, Section 5 lists limitations and next steps, and Appendix A documents every Excel step and formula.

1. Project Overview

Objective, dataset and approach

Objective

Build an interactive HR analytics dashboard that answers three questions: **how big is the attrition problem**, **who is leaving**, and **which factors are associated with leaving**. The output should be easy for an HR manager to read and should end with clear, actionable recommendations.

Dataset

Item	Detail
Source	IBM HR Analytics Employee Attrition and Performance dataset (CSV)
Size	1,470 employee records, 35 original columns
Data quality	No missing values and no duplicate rows
Target field	Attrition (Yes / No): 237 Yes (16.1%) and 1,233 No
Field groups	Demographics, job and role, compensation, satisfaction ratings, tenure and experience, travel and overtime

Tools used

Tool	Used for
Microsoft Excel	Data cleaning, helper columns (bands and labels), validation with PivotTables
Power BI Desktop	Data loading, data model, DAX measures, five-page interactive dashboard
DAX	Attrition rate, retention rate, average income, tenure, and segment-level measures

Approach

- **Prepare:** clean the dataset in Excel, create bands and readable labels, and validate totals with PivotTables.
- **Model:** load the table into Power BI, set data types and sort orders, and create DAX measures.
- **Visualize:** design five pages (summary, attrition drivers, compensation and satisfaction, demographics, insights) with a consistent navy, teal and coral palette.
- **Interpret:** turn the findings into insights and practical recommendations.

2. Data Preparation and Model

From raw CSV to a dashboard-ready table

Columns removed

Column	Reason
EmployeeCount, StandardHours, Over18	Constant value in every row, so they carry no information
DailyRate, HourlyRate, MonthlyRate	No clear business meaning; MonthlyIncome is used for pay analysis instead

Helper columns created in Excel

Column	Purpose
Attrition Flag	1 if Attrition = Yes, else 0, so rates can be calculated with simple DAX
Age Group, Income Band, Tenure Band	Group continuous values into readable bands for charts
Hike Band, Distance Band, Experience Band	Bands for salary hike %, distance from home and total working years
Education Level, Job Satisfaction Level, Env Satisfaction Level, WorkLife Balance Level	Replace numeric codes (1 to 5) with readable labels
Sort columns (one per band)	Keep bands in the correct order in Power BI (Sort by column)

Validation

Before building visuals, PivotTables in Excel were used to confirm: 237 attrition cases, 1,470 employees, an overall rate of 16.1%, and overtime attrition rates of 30.5% (Yes) and 10.4% (No). The same figures were then reproduced by the DAX measures in Power BI.

Key DAX measures

Measure	Definition
Total Employees	`COUNTROWS(HR_Data)`
Attrition Count	`SUM(HR_Data[Attrition Flag])`
Attrition Rate	`DIVIDE([Attrition Count], [Total Employees])`
Retention Rate	`1 - [Attrition Rate]`
Avg Monthly Income	`AVERAGE(HR_Data[MonthlyIncome])`
OT Attrition Rate	`CALCULATE([Attrition Rate], HR_Data[OverTime] = "Yes")`
Early Tenure Attrition Rate	`CALCULATE([Attrition Rate], HR_Data[YearsAtCompany] <= 2)`
High Risk Attrition Rate	`CALCULATE([Attrition Rate], MaritalStatus = "Single", OverTime = "Yes", BusinessTravel = "Travel_Frequently")`

3. Dashboard Walkthrough

Five pages, one consistent design

The dashboard follows a top-down story. It starts with the overall picture, drills into who is leaving, examines pay and satisfaction, profiles the workforce, and closes with insights and recommendations. Every page shares the same title bar, page navigator, color palette and slicers (Department, Gender, Job Role), so filters can be applied consistently.

Design choices

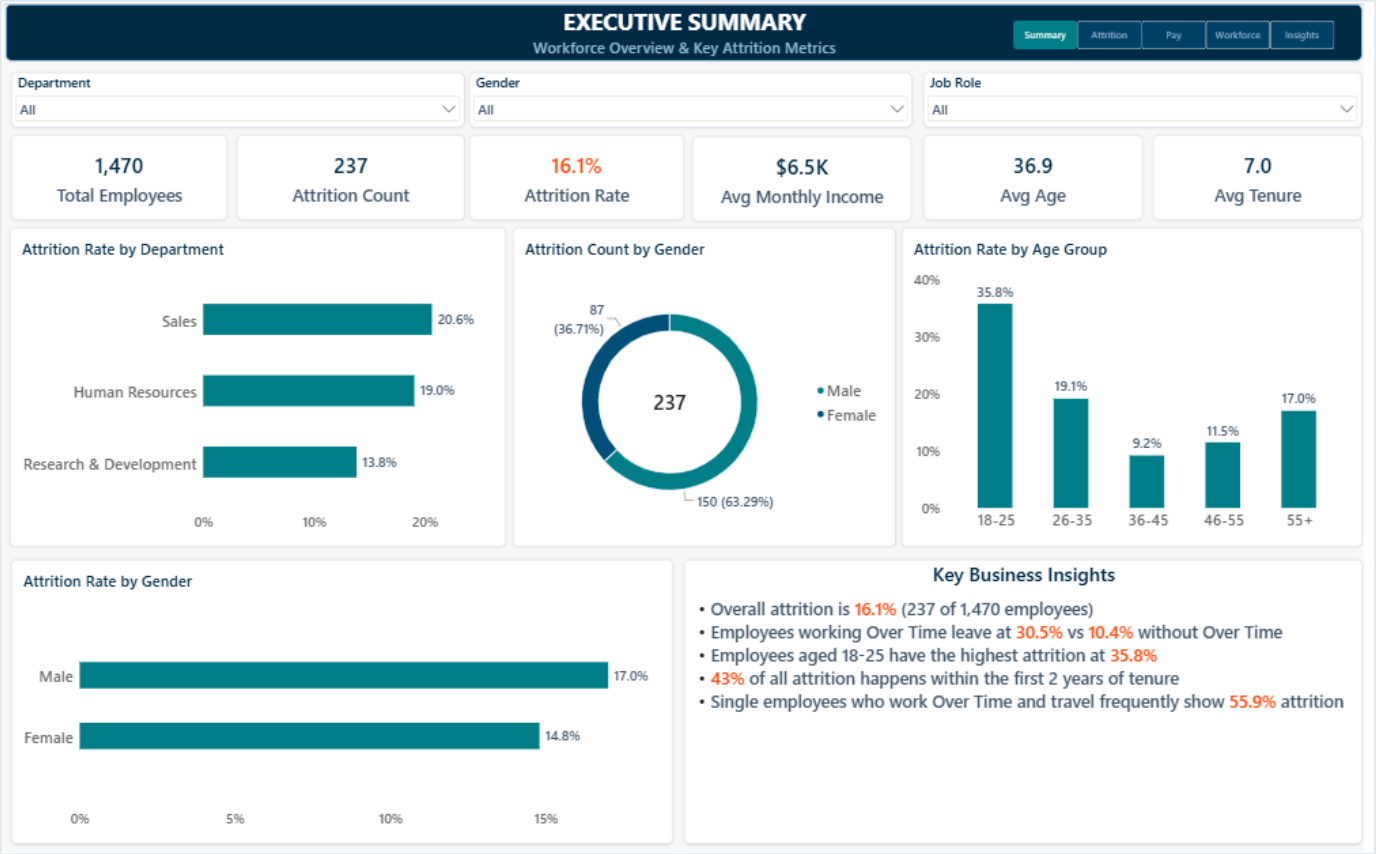
- **Navy** for titles and structure, **teal** for standard values, and **coral** only to highlight problem areas.
- Bands are sorted in logical order (for example 0-2 Yrs to 10+ Yrs), not alphabetically.
- Every chart title states exactly what is measured, and data labels are shown for quick reading.

Dashboard pages

Page	Title	Question answered
1	Executive Summary	How big is the attrition problem overall?
2	Attrition Analysis	Who is leaving: which roles, groups and work conditions?
3	Compensation and Satisfaction	Do pay and satisfaction explain attrition?
4	Workforce Demographics	What does the workforce look like, and where is risk concentrated?
5	Key Insights and Recommendations	What should the organization do about it?

Dashboard Page 1: Executive Summary

Workforce overview and key attrition metrics



What this page shows

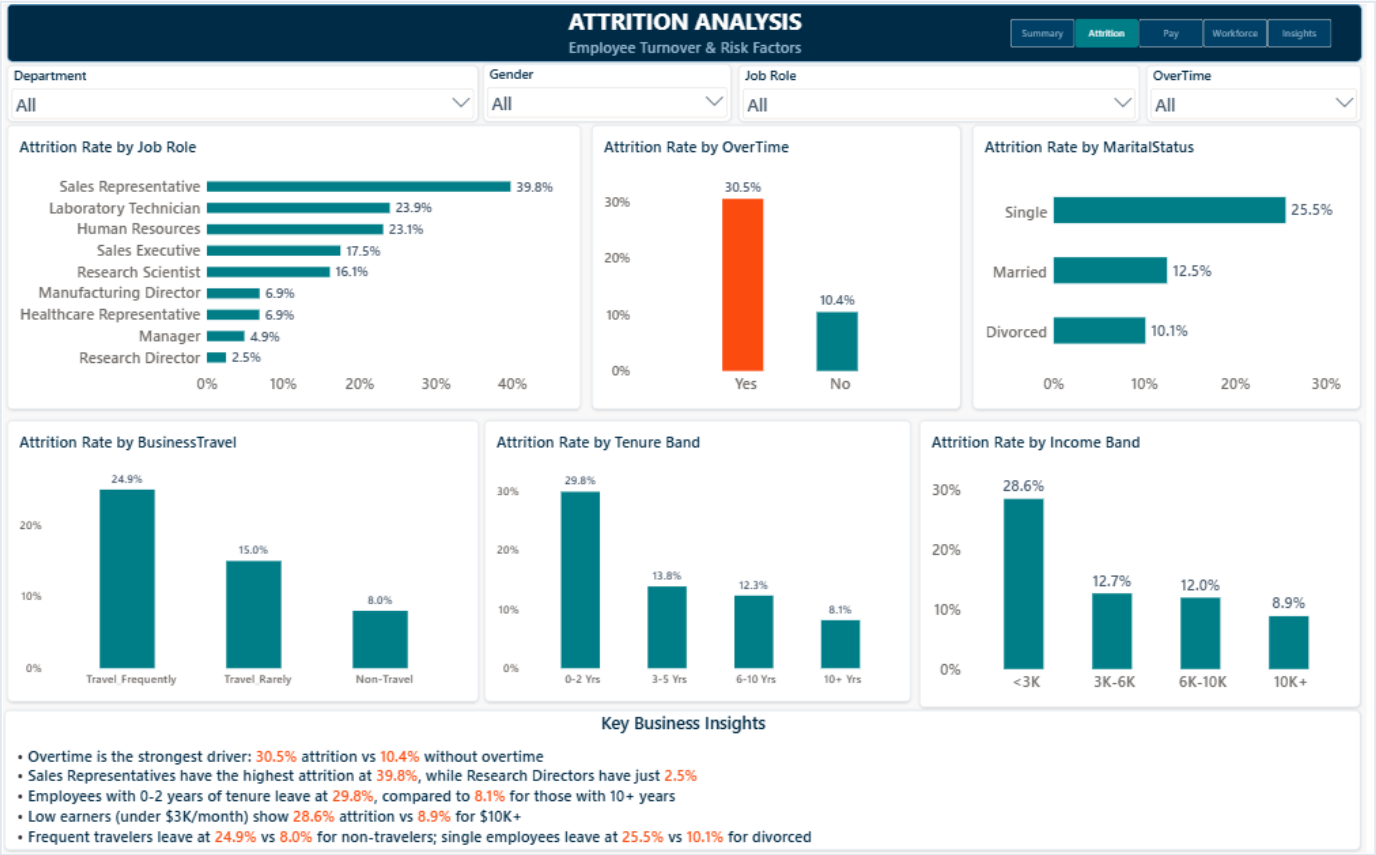
Six KPI cards (total employees, attrition count, attrition rate, average monthly income, average age, average tenure) sit above charts for attrition by department, by gender (count and rate) and by age group. A summary box lists the main insights.

Key findings

- 1,470 employees, 237 left: an overall attrition rate of **16.1%**. Average age is 36.9, average tenure 7.0 years and average monthly income $6.5K.
- Sales has the highest department attrition at **20.6%**, followed by Human Resources (19.0%) and Research and Development (13.8%).
- Attrition is highest among employees aged **18-25 (35.8%)** and falls to 9.2% for ages 36-45.
- Gender has a smaller effect: 17.0% for male and 14.8% for female employees (150 of the 237 leavers are male, 87 are female).

Dashboard Page 2: Attrition Analysis

Employee turnover and risk factors



What this page shows

Six charts break attrition down by job role, overtime, marital status, business travel, tenure band and income band, with slicers for Department, Job Role, Gender and OverTime.

Key findings

- **Overtime:** 30.5% attrition with overtime versus 10.4% without, the largest gap of any single factor.
- **Job role:** Sales Representative (39.8%), Laboratory Technician (23.9%) and Human Resources (23.1%) are the highest; Research Director is the lowest at 2.5%.
- **Marital status:** Single employees leave at 25.5%, versus 12.5% (Married) and 10.1% (Divorced).
- **Travel:** frequent travelers leave at 24.9%, versus 15.0% (rare) and 8.0% (no travel).
- **Tenure and income:** 0-2 years of tenure shows 29.8% attrition (10+ years: 8.1%), and incomes under $3K show 28.6% (above $10K: 8.9%).

Dashboard Page 3: Compensation and Satisfaction

Pay, engagement and retention drivers



What this page shows

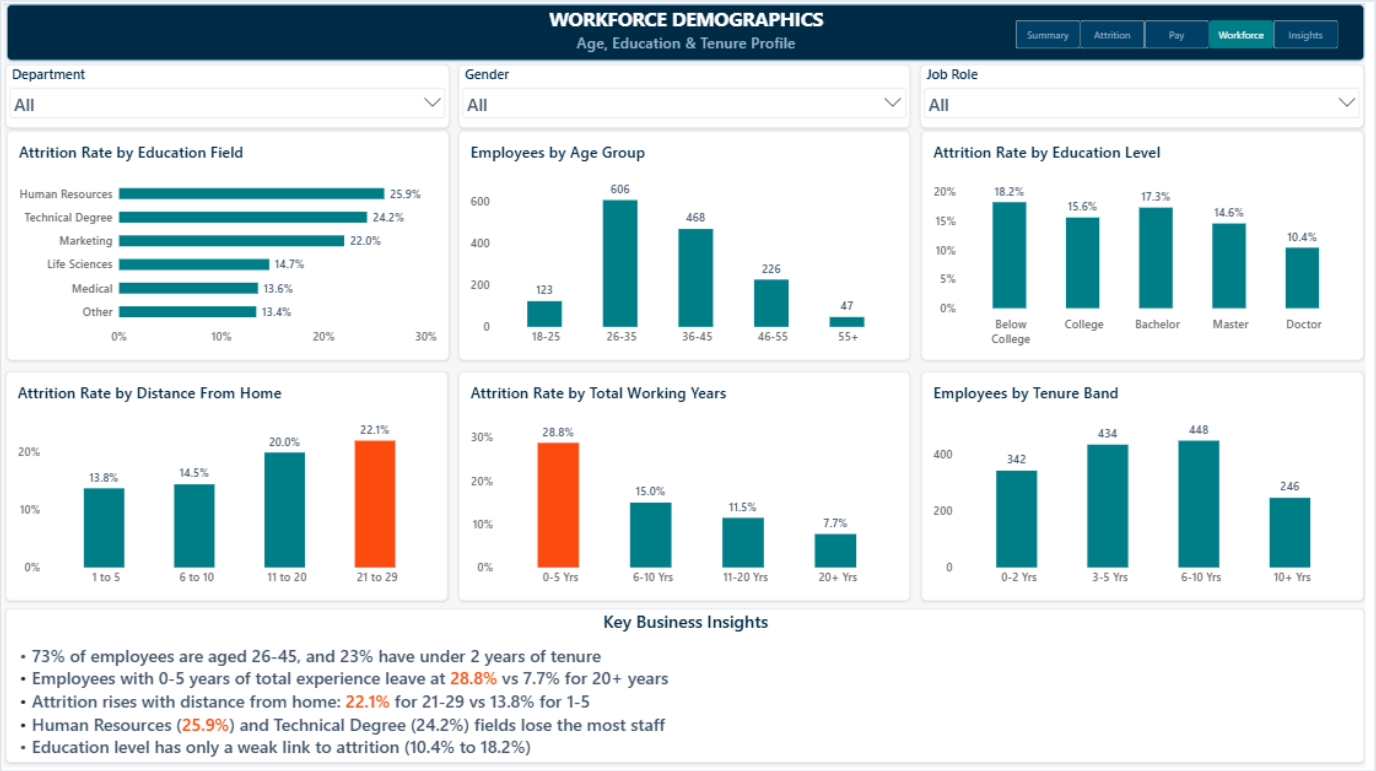
Charts compare average monthly income by job role and by attrition status, and attrition rates by stock option level, salary hike band, job satisfaction and work-life balance.

Key findings

- Employees who left earned **$4.8K** on average versus **$6.8K** for those who stayed (about 30% less).
- Sales Representative has the lowest average income ($2.6K) and also the highest attrition, highlighted in coral.
- **Stock options:** 24.4% attrition with no stock options versus 9.4% at level 1 and 7.6% at level 2. Level 3 is a small group, so its 17.6% should be read with caution.
- **Salary hike %:** attrition stays between 14.6% and 17.5% across bands, so there is no clear link.
- **Satisfaction:** low job satisfaction shows 22.8% attrition versus 11.3% for very high. Bad work-life balance shows the highest attrition at **31.3%**.

Dashboard Page 4: Workforce Demographics

Age, education and tenure profile



What this page shows

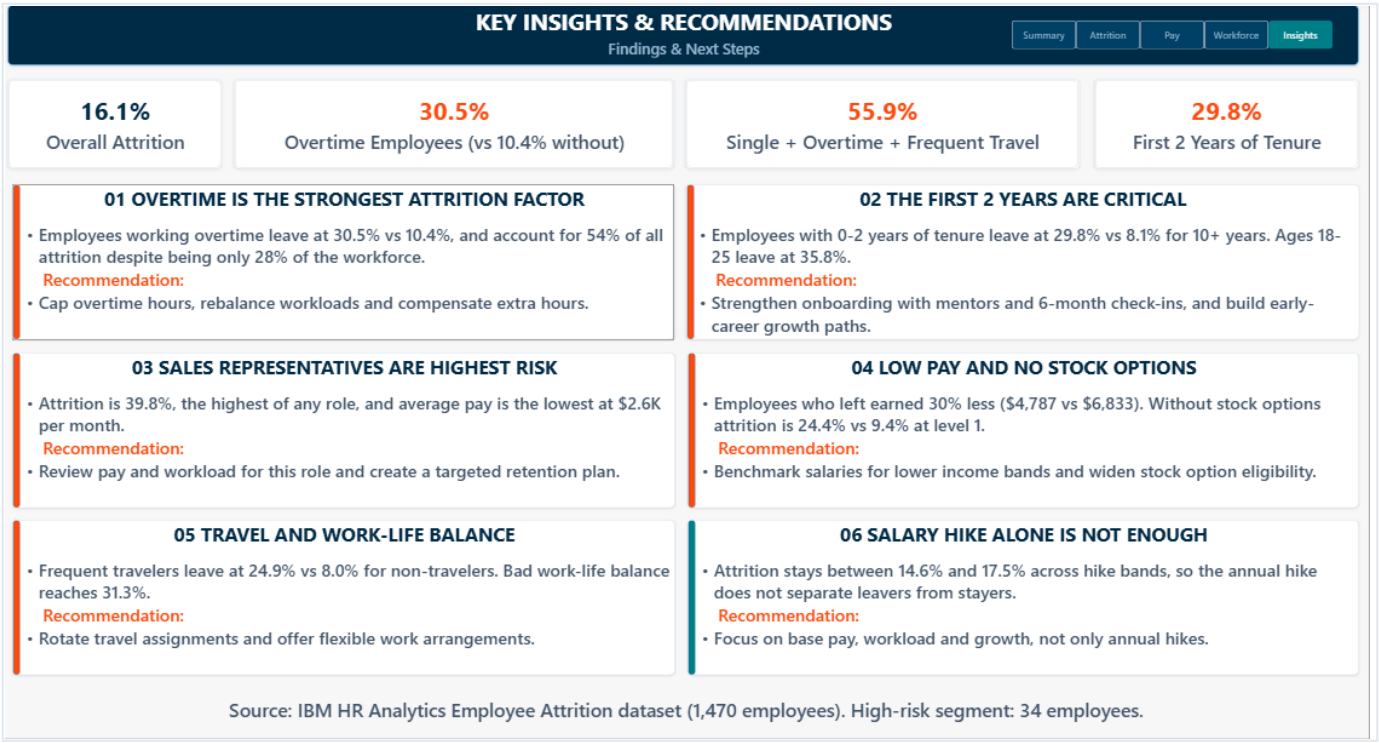
Charts show attrition by education field and level, headcount by age group and tenure band, and attrition by distance from home and total working years.

Key findings

- About **73%** of employees are aged 26-45, and about 23% have under 2 years of tenure (342 employees).
- Employees with 0-5 years of total experience leave at **28.8%**, versus 7.7% for those with 20+ years.
- Attrition rises with distance from home: 13.8% for the closest band and **22.1%** for the farthest (21 to 29).
- Education field: Human Resources (25.9%) and Technical Degree (24.2%) are highest, but Human Resources is a very small group (27 employees).
- Education level has only a weak link to attrition, ranging from 10.4% (Doctor) to 18.2% (Below College).

Dashboard Page 5: Key Insights

Findings and next steps



What this page shows

Four headline metrics and six insight cards, each with supporting evidence and a recommended action, summarize the analysis for a decision-maker. This page is not filtered by slicers and always reflects the full dataset.

Key findings

- Overtime, early tenure, the Sales Representative role, low pay with no stock options, and travel with poor work-life balance are the main risk areas.
- Employees who are **single, work overtime and travel frequently** show 55.9% attrition (19 of 34), about 3.5 times the company average. This is a small group and should be treated as a signal rather than a precise rate.
- Salary hike alone is not enough: base pay, workload and growth appear to matter more.

4. Insights and Recommendations

What to act on first

#	Insight	Evidence	Recommendation
1	**Overtime has the strongest link to attrition**	30.5% vs 10.4%. Overtime employees are 28% of staff but 54% of leavers.	Cap overtime hours, rebalance workloads and compensate extra hours.
2	**The first 2 years are critical**	29.8% for 0-2 years vs 8.1% for 10+ years. Ages 18-25 leave at 35.8%.	Strengthen onboarding with mentors and 6-month check-ins; build early-career growth paths.
3	**Sales Representatives are highest risk**	39.8% attrition and the lowest average pay ($2.6K per month).	Review pay and workload for this role and create a targeted retention plan.
4	**Low pay and no stock options**	Leavers earned 30% less ($4,787 vs $6,833). No stock options: 24.4% vs 9.4% at level 1.	Benchmark salaries for lower income bands and widen stock option eligibility.
5	**Travel and work-life balance**	Frequent travel: 24.9% vs 8.0% for non-travelers. Bad work-life balance: 31.3%.	Rotate travel assignments and offer flexible work arrangements.
6	**Salary hike alone is not enough**	14.6% to 17.5% attrition across hike bands; no clear pattern.	Focus on base pay, workload and growth, not only annual hikes.

Priority segment

Employees who are single, work overtime and travel frequently have a 55.9% attrition rate (19 of 34 employees). Even with a small group, this combination is the clearest place to start a pilot retention program, because three separate risk factors overlap.

5. Limitations and Next Steps

Reading the results responsibly

Limitations

- **Association, not causation.** The dashboard shows which factors are linked to attrition. It cannot prove that overtime, pay or travel caused employees to leave.
- **Fictional dataset.** The IBM HR Analytics dataset was created for illustration, so the findings demonstrate analytical method rather than conclusions about a real company.
- **Small groups.** Some segments are small (for example the 34-employee high-risk group, Human Resources education field with 27 employees, and stock option level 3), so their rates can swing widely.
- **Snapshot data.** There are no dates, so trends over time and time-to-exit cannot be analyzed.
- **No statistical testing.** Differences are descriptive; significance tests and multivariate models were not used.

Next steps

- Test the strongest factors together (for example logistic regression) to see which remain important after accounting for the others.
- Add an attrition risk score or segment flag to identify at-risk employees earlier.
- Add a drill-through page for each job role with an employee-level view.
- Publish to Power BI Service so stakeholders can explore the dashboard interactively.

Conclusion

Attrition in this dataset is concentrated rather than evenly spread. A small number of conditions, overtime, early tenure, low pay in specific roles, frequent travel and poor work-life balance, account for a large share of departures. Targeting these areas first is likely to deliver the largest retention benefit for the least effort.

Syed Ubedullah Basha
Data Analyst | Power BI Developer | SQL Developer
syedubedullahbasha@gmail.com | linkedin.com/in/syed-ubedullah-basha | github.com/syed-ubedullah

Appendix A: Excel Steps and Formulas

Everything done in Excel before loading the data into Power BI

Workflow

Step	What was done	Result
1	Opened the CSV in Excel and saved it as an Excel workbook	IBM HR Analytics Dashboard project.xlsx, sheet HR_Data
2	Converted the data range into an Excel Table (Ctrl + T) and named it HR_Data	1,470 rows, 35 columns
3	Deleted 6 unneeded columns (EmployeeCount, StandardHours, Over18, DailyRate, HourlyRate, MonthlyRate)	29 columns
4	Added 17 helper columns (bands, labels, flag and sort columns)	46 columns
5	Validated with PivotTables: 237 attrition cases, 1,470 employees, 16.1% overall, overtime 30.5% vs 10.4%, and no blank bands	Data and formulas confirmed
6	Wrote Distance Band labels as '1 to 5' instead of '1-5' because Excel read values like '6-10' as dates	Warning removed, column stays text

Helper column formulas

Column	Formula
Attrition Flag	`=IF([@Attrition]="Yes",1,0)`
Age Group	`=IF([@Age]<=25,"18-25",IF([@Age]<=35,"26-35",IF([@Age]<=45,"36-45",IF([@Age]<=55,"46-55","55+"))))`
Income Band	`=IF([@MonthlyIncome]<3000,"<3K",IF([@MonthlyIncome]<6000,"3K-6K",IF([@MonthlyIncome]<10000,"6K-10K","10K+")))`
Tenure Band	`=IF([@YearsAtCompany]<=2,"0-2 Yrs",IF([@YearsAtCompany]<=5,"3-5 Yrs",IF([@YearsAtCompany]<=10,"6-10 Yrs","10+ Yrs")))`
Hike Band	`=IF([@PercentSalaryHike]<=13,"11-13%",IF([@PercentSalaryHike]<=16,"14-16%",IF([@PercentSalaryHike]<=19,"17-19%","20-25%")))`
Distance Band	`=IF([@DistanceFromHome]<=5,"1 to 5",IF([@DistanceFromHome]<=10,"6 to 10",IF([@DistanceFromHome]<=20,"11 to 20","21 to 29")))`
Experience Band	`=IF([@TotalWorkingYears]<=5,"0-5 Yrs",IF([@TotalWorkingYears]<=10,"6-10 Yrs",IF([@TotalWorkingYears]<=20,"11-20 Yrs","20+ Yrs")))`
Education Level	`=CHOOSE([@Education],"Below College","College","Bachelor","Master","Doctor")`
Job Satisfaction Level	`=CHOOSE([@JobSatisfaction],"Low","Medium","High","Very High")`
Env Satisfaction Level	`=CHOOSE([@EnvironmentSatisfaction],"Low","Medium","High","Very High")`
WorkLife Balance Level	`=CHOOSE([@WorkLifeBalance],"Bad","Good","Better","Best")`
Sort columns (Age, Income, Tenure, Hike, Distance, Experience)	Same IF logic as the matching band, returning 1, 2, 3, 4 (or 5 for Age Group) so Power BI can sort bands in order